

AVNISH PATEL

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PUBLICATION [\[LINK\]](#)

- B. Maxwell, A. Patel, "Logarithmic Lenses: Exploring Log RGB Data for Image Classification", CVPR 2024

EXPERIENCE

Medtronic | Robotics Software Engineer, Boston

Jan. 2024 – Apr. 2025 | Sept. 2025 - Present

Ground Truth Generation (Python) with Camera Calibration & Deep Learning (PyTorch)

- Built an end-to-end *SLAM* pipeline using *stereo-rectified*, semantically segmented images in *DROID-SLAM* for *dense depth estimation* in surgical videos and scene understanding of anatomy and surgical tools
- Optimized *camera trajectory* using *GTSAM Factor Graphs non-linear optimization* and refining *3D reconstruction* with *Bundle Adjustment* and *Multi-View Geometry* with *LightGlue feature matching* and *Open3D offline rendering pipeline*
- Designed a *Pose Graph* pipeline with *GTSAM* using *Robot Kinematics* and *TensorRT* based *YOLOv8* vision keypoints to reduce the drift and track the instrument points accurately in *Out-of-View* scenarios
- Engineered a *Arm Collision Avoidance System* using *Trimesh* and *Pytorch Volumetric Signed Distance Fields* of the medical robot using *URDF* and *Kinematics* using *RTI DDS publish-subscribe* method for rendering a *Digital Twin* of the robot
- Deployed a *C++* based *3D registration pipeline* on *NVIDIA Holoscan* using *Fast Foundation Stereo* and *UniMatch*, with custom *CUDA* kernels for high-performance *GPU* acceleration
- Developed a *monocular depth estimation* pipeline using *Depth Anything* to estimate real-time distances between surgical instrument tips, enabling spatial awareness and precise instrument interaction analysis
- Built a *MASt3R SLAM*-based registration pipeline to create a *3D Digital Twin* of surgical robot carts using *GTSAM* for multi-camera scale correction and pose optimization, and *Signed Distance Fields (SDFs)* for collision avoidance.
- Programmed a *stereo-vision-based localization* system using *ORB-SLAM3* and *RealSense D455* cameras, enabling accurate mapping of *surgical cart trajectories* in a medical environment
- Developed a real-time *Ground Truth pose estimation pipeline* using *OptiTrack* camera capture and *robot kinematics* with *PnP* and *ROMA feature detection* for training deep learning models on *instrument articulation*
- Automated *Endoscope Stereo Camera Calibration* using *Zhang's method* on a *Charuco* board with a *UR3 Robot Arm*
- Segmented hernias in 10,000 medical images using *Swin Base Transformer* with *PyTorch DDP* for *multi-GPU training*, tracked experiments via *MLflow*, and optimized using *ONNX/TensorRT*
- Implemented a *PyTorch* wrapper with *Optical Flow* on *FAST API* using *Unimatch*, deploying models to *ONNX/TensorRT* for 10x reduction in real time annotation of medical image frames with 1-second latency

Visiting Researcher - AirLab | [Report](#) | Carnegie Mellon University, Remote

July 2025 – May 2026

- Integrated *Relative Pose Graph Optimization* in *ROS2* in *C++* using *GTSAM Factor Graphs Fixed-Lag Smoother* in the *IMU Preintegration* module of the *multi-modal IMU-LiDAR sensor fusion* to reduce long term drift in *SLAM*
- Executed trajectory mapping using *Livox LiDAR* and *IMU* sensors on the *Unitree G1 Humanoid* robot, applying a *low-pass filter* to mitigate *IMU bias* and enhance mapping accuracy
- Developed a *multi-modal sensor fusion* visual front-end for *Visual-Inertial Odometry* based on *VINS Mono* with *learned keypoint* detection and *patch-based optical flow* tracking across *Monocular, Stereo, and Thermal* camera configurations, providing uncertainty-aware measurements for downstream optimization
- Built a *tightly-coupled VIO* back-end with *Ceres* in *C++* using *Variance Component Analysis (VCA)* to adaptively weight *Camera* and *IMU* residuals within a *sliding-window Bundle Adjustment* framework to dynamically balance sensor contributions under degraded conditions
- Evaluated the *VIO* pipeline on multiple datasets like *Euroc, CART, SubT-MRS* datasets on challenging and degrading conditions like smoke, dust and low texture environments

Research Student – Computer Vision Lab | Northeastern University, Boston

May. 2023 – Dec. 2023

- Researched *Raw Log RGB* data's impact on deep networks like *ResNet-18*, improving classification performance and robustness to intensity and color variations, on novel *RAW10* dataset (10K DNG & JPG images each, 10 categories)

Kisan Drip Irrigation Ltd. | Artificial Intelligence Engineer, India

Aug. 2020 – Aug. 2022

- Integrated *ElasticFusion: RGB-D SLAM* with *C++* to align *multi-view point clouds* from *Intel RealSense D455* cameras, enabling accurate *3D Reconstruction* for *pipe inspection* and *defect analysis*
- Experimented with *PointNet-based* deep learning models in *Python* for *point cloud classification* to enhance complex defect identification, achieving a 30% improvement over traditional 2D vision methods
- Implemented the *YOLO-based object detection* to localize drippers in pipe assemblies, enabling precise hole punching
- Deployed the *YOLO pipeline* as a *Docker containerized FastAPI* service integrated into manufacturing workflows

EDUCATION

Northeastern University

Masters of Science in Robotics, ECE Concentration

Boston, MA

Sep. 2022 – Dec. 2024